

Practical Medical chemistry and laboratory Safety



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1st year Medical Students

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Introduction

What is the chemistry?

Chemistry is a science, which study the properties, composition and structure of the substances and all obtaining energy changes (endothermic and exothermic).

Dangerous substances:-

The dangerous substances are:-

1- Acids: H_2SO_4 , HCl , HNO_3

2- Bases: NaOH , NH_4OH

3- Organic solvents: benzene (C_6H_6), chloroform (CHCl_3) , Aniline ($\text{C}_6\text{H}_5\text{NH}_2$) , phenol ($\text{C}_6\text{H}_5\text{OH}$) , Toluene ($\text{C}_6\text{H}_5\text{CH}_3$) , Methanol , Ethanol ,etc.

Essential rules for laboratory safety:-

Always:

1. Familiarize yourself with laboratory safety procedures.
2. Dress apron.
3. Read the procedure before starting any experiment.
4. Check that the apparatus assembled correctly.
5. Handle all chemicals with great care.
6. Keep your working area tidy.
7. Check to see that the gas and water taps are turned off.
8. Ask your instructor if in doubt.
9. Wash your hands before leaving the laboratory.

Never:

1. Eat or drink in the laboratory.
2. Smoke in the laboratory.
3. Inhale or taste chemicals.
4. Working alone.
5. Play mad scientist in the laboratory.

Glass ware and equipments used in the laboratory:-

1. **Beaker:** to store, mix, and heat liquids.
2. **Conical flask:** for titration and collection of solution.
3. **Burette:** to measure the volume of a liquid.
4. **Graduated pipette:** for accurate measurement of liquid.
5. **Burette stand:** to hold and stabilize burette during titration .
6. **Clamp:** for holding burette, test tube, beaker and more.
7. **Funnel:** to direct liquids into containers that have small openings.
8. **Test tube:** to hold, mix, or heat small quantities of solid or liquid chemicals.
9. **Dropper:** for measuring and transferring liquids in small amounts.

10. **Graduated cylinder:** for measuring liquids.
11. **Brush:** for cleaning the glass ware.
12. **Volumetric flask:** for preparation of accurate concentration.
13. **Spatula:** for transfer of solid.
14. **Washing bottle:** for washing the glass.
15. **Tongs:** for holding test tube.
16. **Filter paper:** for the separation of solid particles from liquids.
17. **Watch glass:** for weighting of solids.
18. **Glass rod:** for stirring the solution.
19. **Test tube rack:** for collecting of the test tube.

Common Laboratory Equipment



burette

clamp

measuring
marks

tap

stand

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Chemical Analysis:

Chemical analysis is basically depending on the determination of chemical composition for any substances, and can be divided into two parts:-

1- Qualitative analysis:-

Qualitative analysis shows the containing elements or (ions) for any substance.

2- Quantitative analysis:-

Quantitative analysis determines the amounts of element which present in the substance.

Molarity (M):- The number of moles of solute present in one
litter of solution.

$$M = \text{mol/L}$$

$$M = \frac{\text{Wt of solute} \times 1000}{\text{M.Wt of solute} \times V_{\text{solution}}}$$

Example:- what is the molar concentration of a 0.85% (w/v)
solution of NaCl (M.Wt = 58.4 g/mol)?

Since % (w/v) represent grams per 100ml, there for

$$\begin{aligned} M &= 0.85 \times 1000 / 100 \times 58.4 \\ &= 0.146 \text{ M (mol/L)} \end{aligned}$$

Normality (N):- The number of gram equivalent weight of solute per one liter of solution.

$$N = \text{equiv/L}$$

$$N = \text{Wt of solute} \times 1000 / \text{Eq.Wt of solute} \times V_{\text{solution}}$$

Example:- How many grams of Na_2SO_4 ((M.Wt = 142 g/mol ,
Eq.wt = 71 g/eq.Wt)are required to prepare 200ml of a 0.5N
solution?

$$\begin{aligned}\text{Wt of } \text{Na}_2\text{SO}_4 &= (N \times V \times \text{eq.Wt}) / 1000 \\ &= (0.5 \times 200 \times 71) / 1000 \\ &= 7.1 \text{ g}\end{aligned}$$

Solid substances:-

Amount of solid in grams (g) = $(N \times V \times \text{eq.Wt}) / 1000$

eq.Wt = equivalent weight = molecular weight / n

n = No. of (H⁺) in case of acid,
(-OH) in case of base or
(e⁻) in case of redox.

Liquid substances:-

The conc. of the acid in the bottle can be found by the following reaction:-

$$N = (\text{specific gravity} \times \% \times 1000) / \text{eq.wt}$$

Many standard solutions can not be prepared accurately by the direct method for example:

- 1- Sodium hydroxide solution is made from a solid that rapidly pick up moisture.
- 2- Solution of strong acids like hydrochloric acid and sulphuric acid which are used widely for preparation of standard solution, they are volatile when prepared. In this case a solution of approximately desired concentration can be prepared and then standerized by another suitable standard solution.

For preparation of standard solution the substance should have the following properties:-

- 1- Dry.
- 2- Pure.
- 3- Non hygroscopic nor efflorescent.
- 4- Do not affected by O₂ and CO₂ and have high equivalent weight.
- 5- Stable during storage.
- 6- Easily obtained and favorably cheap.